AwTherm
Water Activity Meter

AIR CHIP TECHNOLOGY



STABLE TEMPERATURE YIELDS PRECISE MEASUREMENTS.

- ROTRONIC HYGROMER® IN-1
- Highest precision through exact stabilization of temperature
- Wide temperature-control range
- Excellent reproducibility
- Excellent long-term stability
- Variable size of sample container
- Reference probe exchangeable for calibration or cleaning
- Quickest measurements with the AwQuick function
- Simple operation via Touch Interface
- Conforms to ISO 21807 / EMC 2008/108/EC / IEC EN 61010-1:2010



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MEASUREMENT SOLUTIONS

TECHNICAL INFORMATION

Temperature-stabilized measurement

With AwTherm, ROTRONIC offers a professional, high-end laboratory unit for temperature-stabilized measurement of water activity in the foodstuffs, cosmetics, or pharmaceutical industries.

The wide control range permits measurements to be integrated directly in the tempered manufacturing or storage process. Water-activity measurement reacts very sensitively to variances in temperature. Stabilizing the temperature prevents imprecise results due to external temperature influences.

Exchangeable measuring head

A further advantage of AwTherm is that the measuring head is removable, which means that the probe can be temperature-calibrated or adjusted to give high precision.

User-friendliness

The AwTherm shows its paces in island-operation mode, with its simple handling and clarity. No laboratory should be without the device in conjunction with the ROTRONIC HW4 software

Why measure water activity?

The free water in a product influences its microbiological, chemical and enzymatic stability. If there is too much free water present, the products spoil; too little water can be detrimental to the product characteristics.

Water activity	Contaminants
aw = 0.91..0.95	Many bacteria
aw = 0.88	Many yeasts
aw = 0.80	Many mildews
aw = 0.75	Halophile bacteria
aw = 0.70	Osmiophile yeasts
aw = 0.65	Xerophile mildew

Measuring water activity provides useful information about characteristics such as cohesion, storage lifetime, clotting or pourability of powders, tablets, etc., or the adhesive properties of coatings.

General	
Device type	Temperature-stabilized water-activity measurement
Operating conditions	1...40 °C (34...107 °F)
Measurement range	0.005...1.000 aw
Accuracy	±0,005 aw (10...30 °C) ±0.1 °C (±0.18 °F)
Power supply	110..230 V / 50..60 Hz
Display	8-line LCD with touch operation
Sample sizes	Variable (14 mm / 40 mm)
Technical Information	
Current Consumption	≤2 A
Temperature-control range	0...60 °C (32...140 °F)
Temperature stability	±0.01 °C/min (±0.018 °F/min)
Chamber-temperature gradient	‹0.1 °C (‹0.18 °F)
Firmware update	Via USB port
Functions	
HW4-compatible	Yes (v 3.6.0 and higher)
aw-Quick function	Yes
Interface	Micro USB
Calibration / Adjustment	Aw & Temperature: HG2-S (probe removed with AwT-CAL) Aw: ROTRONIC humidity standards (via HW4)
Trend indicator	Yes
Probes	
Sensor	HYGROMER® IN-1
Maintenance / Calibration	Annual calibration (recommended)
Long-Term Stability	‹0.01 aw/year
Temperature sensor	Pt100, DIN 1/3 Class B
Approval / Conformity	
Standards	ISO 21807
CE / EMC	EMC 2004/108/EC IEC EN 61010-1:2010
IP protection	IP21
Housing / Mechanical parts	
Housing material	PC / ABS
Housing dimensions	400 x 180 x 180 mm
Sample container dimensions	AwT-PS14: Ø46 x 14 mm AwT-PS40: Ø46 x 40 mm
Weight	4200 g

Subject to technical change without notice. Printing and other errors reserved.

Would you like more information?

You will find a complete and up-to-date overview of the products on our Internet page at **www.rotronic.com**.

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ROTRONIC AG, Grindelstrasse 6, CH-8303 Bassersdorf, Tel. +41 44 838 11 44, www.rotronic.ch
ROTRONIC Instruments (UK) Ltd, Crompton Fields, Crompton Way, Crawley, West Sussex, RH10 9EE, UK, Phone +44 (0)1293 571000, www.rotronic.co.uk
ROTRONIC Instrument Corp, 135 Engineers Road, Hauppauge, NY 11788, USA , Phone, +1 631 427-3898, www.rotronic-usa.com
ROTRONIC Canada Inc., 236 Pritchard Rd, Unit 204, Hamilton, ON, Canada , L8W 3P7, Phone + 1 416-848-7524, www.rotronic.ca
ROTRONIC Instruments Pte. Ltd., 1003 Bukit Merah Central, #06-31 Inno Centre, Singapore 159836, Phone +65 6376 2107, www.rotronic.sg

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